

Appendix A

Constellation Orbital Mechanics

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This appendix presents the complete orbital mechanics analysis for the HELIOS three-node constellation. Section A.1 introduces the Sun–Earth CR3BP computational framework and normalization constants. Section A.2 provides the full trajectory analysis for the L1 flagship node, including differential correction, stability, Solar Exclusion Zone (SEZ) compliance, and Solar Radiation Pressure (SRP) perturbation. Section A.3 summarizes the L4/L5 node orbital regime. Section A.4 consolidates the propellant and ΔV budget for the full constellation.

A.1 Computational Framework

All L1 trajectory computations are performed in the Sun–Earth Circular Restricted Three-Body Problem (CR3BP) rotating frame, normalized by $L^* = 1.496 \times 10^8$ km and $T^* = 1/n_{Earth}$. The Sun is fixed at $x = -\mu$, the Earth at $x = 1 - \mu$, with $\mu = 3.040357 \times 10^{-6}$ (JPL DE440, lunar mass absorbed). Equations of motion are integrated with SciPy RK45 at $rtol = 10^{-12}$, $atol = 10^{-14}$; measured Jacobi-constant drift over one full period is $|\Delta C| = 4.4 \times 10^{-16}$, confirming machine-level fidelity. All constants are listed in Table A.1.

Halo orbit initial conditions are generated from the third-order Richardson (1980) [1] expansion and refined by a single-shooting differential corrector (Howell 1984 [2]), simultaneously adjusting x_0 and $\dot{y}_0$ via a 2×2 STM-based Newton iteration until $|vx(T/2)| < 10^{-10}$ and $|vz(T/2)| < 10^{-10}$. The corrector is seeded with an Az-targeting outer loop that scales z_0 iteratively until the measured out-of-plane amplitude of the converged periodic orbit matches the 250,000 km design target within 200 km. Orbit stability is characterized via the monodromy matrix $\Phi(T, 0)$, obtained by propagating the State Transition Matrix over one full period alongside the nominal trajectory. The L4/L5 budget is sized from published station-keeping rates for Jupiter-perturbed tadpole-libration orbits — differential correction is unnecessary, as the triangular equilibria are linearly stable in the Sun–Earth mass ratio.

Parameter	Value	Notes / Derivation
Mass parameter μ	3.040357×10^{-6}	Sun–Earth; Moon mass absorbed into Earth (JPL DE440)
Length unit L^*	1.496×10^8 km	1 AU, Sun–Earth mean semi-major axis
Time unit T^*	5.022548×10^6 s (58.131 d)	$T^* = 1 / n_{Earth} = T_{Earth} / (2\pi)$
Velocity unit V^*	29.786 km/s	$V^* = L^* / T^*$ ($\approx$ Earth mean orbital velocity)
Integrator	RK45 (SciPy solve_ivp)	$rtol = 10^{-12}$, $atol = 10^{-14}$
L1 equilibrium x_{L1}	0.989986 (norm.)	Newton–Raphson on quintic; L1–Earth: 1,497,631 km; L1–Sun: 1.481×10^8 km

Table A.1 — CR3BP Sun–Earth computational constants. All values used directly by the numerical orbit propagator.

A.2 L1 Node — Northern Halo Orbit

A.2.1 Design Rationale and Heritage

The L1 halo orbit was selected to satisfy three mission-critical requirements simultaneously: uninterrupted upstream CME and solar wind monitoring; zero violations of the $\pm 5^\circ$ Solar Exclusion Zone communication threshold; and low-risk insertion using a stable-manifold trajectory. The Northern halo family ($m = +1$ in Richardson notation) was chosen to maintain a consistent coronagraph reference frame; the designation refers to the bifurcation branch from the planar Lyapunov orbit, not a constraint on z-sign throughout the trajectory.

The out-of-plane amplitude $A_z \approx 250,000$ km was selected as the primary design point. This amplitude places the spacecraft at a minimum angular separation of 9.51° from the Sun–Earth line — exceeding the $\pm 5^\circ$ SEZ requirement with a 4.51° margin — while remaining well within the range validated by operational missions. The large $A_y = 705,322$ km amplitude enables stable-manifold insertion within a total budget of ≤ 50 m/s, consistent with the manifold approach validated by the Genesis mission (Lo et al. [3]).

Mission	Orbit Type	A_z (km)
ACE (1997 – present)	L1 Lissajous	$\sim 157,000$
SOHO (1995 – present)	L1 Halo	$\sim 120,000$
Genesis (2001 – 2004)	L1 Halo	$\sim 300,000$
DSCOVR (2015 – present)	L1 Lissajous	$\sim 160,000$
HELIOS L1 (Design)	L1 Northern Halo	$\sim 250,000$

Table A.2 — Heritage mission comparison for Sun–Earth L1 orbital parameters.

A.2.2 Computed Orbital Parameters

Table A.3 presents the complete parameter set for the differentially corrected Northern halo orbit at $A_z \approx 250,000$ km. The corrector converged to $|v_x(T/2)| < 10^{-10}$ and $|v_z(T/2)| < 10^{-10}$ in normalized units, confirming machine-level periodicity. The measured in-plane amplitude $A_x = 220,908$ km substantially exceeds the Richardson third-order estimate of $\sim 82,500$ km — this is physically expected, as the perturbation expansion underestimates the nonlinear x-excursion at large amplitudes; the value in Table A.3 is derived from the full nonlinear propagation and is the operationally relevant quantity.

Parameter	Value	Notes
Orbit type	Northern L1 Halo	Richardson [1] + DC [2]
A_z (out-of-plane)	249,993 km	Design target 250,000 km
A_y (cross-track)	705,322 km	Supports manifold insertion
A_x (radial)	220,908 km	Nonlinear propagation value
Period T	177.74 days	DC-converged (≈ 5.9 months)
Jacobi constant C	3.00080980	$ 4C < 10^{-15}$ verified

SEZ clearance	9.51° minimum	Exceeds $\pm 5^\circ$ requirement
Eclipse fraction	0.00%	L1 geometry precludes eclipse
Stability index ν	804.0	Unstable; SK required
Divergence timescale τ	24.1 days	Without station-keeping
SRP ΔV	3.58 m/s/yr	Included in SK budget

Table A.3 — Computed orbital parameters for the HELIOS L1 Northern halo orbit ($A_z \approx 250,000$ km).

A.2.3 Stability and Perturbation Environment

The monodromy matrix eigenvalue analysis yields a stability index $\nu = 804.0$, confirming the orbit is dynamically unstable with a characteristic divergence timescale of 24.1 days. This mandates continuous station-keeping via Floquet-mode [4] targeting at a baseline rate of 4.0 m/s/yr, consistent with Genesis operational experience (3.8 m/s/yr measured [3]).

Solar Radiation Pressure is the dominant non-gravitational perturbation; for the L1 spacecraft ($C_r = 1.5, A/m = 0.0166 \text{ m}^2/\text{kg}$), the SRP acceleration of $1.13 \times 10^{-7} \text{ m/s}^2$ produces an uncompensated position drift of 433 km over 30 days, growing to 434,000 km over one period due to manifold amplification. The resulting 3.6 m/s/yr SRP contribution is absorbed within the 60 m/s station-keeping allocation. Figures A.1–A.5 illustrate the orbit geometry, projections, SEZ compliance, SRP effect, and ΔV budget.

A.2.4 Governing Equations

The synodic (rotating) frame co-rotates with the Sun–Earth line: $\hat{x}$ points Sun→Earth, $\hat{z}$ is directed along the ecliptic angular momentum (north), and $\hat{y} = \hat{z} \times \hat{x}$ completes the right-handed system. All coordinates are nondimensionalized by L^* and T^* (Table A.1).

The CR3BP equations of motion in the rotating frame are:

$$\ddot{x} - 2\dot{y} = \partial\Omega/\partial x, \quad \ddot{y} + 2\dot{x} = \partial\Omega/\partial y, \quad \ddot{z} = \partial\Omega/\partial z$$

where $\Omega = \frac{1}{2}(x^2 + y^2) + (1 - \mu)/r_1 + \mu/r_2$ is the effective pseudo-potential, r_1 and r_2 are distances to the Sun and Earth respectively, and $\mu = 3.040357 \times 10^{-6}$ is the mass parameter (Moon mass lumped with Earth, JPL DE440). The Jacobi constant — the sole integral of motion in the CR3BP — is:

$$C = 2\Omega - (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = 3.00080980$$

Measured drift over one full period is $|\Delta C| < 10^{-15}$, confirming integration fidelity at machine precision.

Initial conditions are generated via Richardson's (1980) third-order Lindstedt–Poincaré approximation, then refined to a numerically exact periodic orbit using Howell's (1984) single-shooting differential corrector with A_z -amplitude targeting. The monodromy matrix yields stability index $\nu = 804$, corresponding to a divergence timescale $\tau_{\text{div}} = T/\ln \nu \approx 24$ days and necessitating active station-keeping (4 m/s/yr allocated).

A.3 L4/L5 Nodes — Tadpole Libration Orbits

The Sun–Earth triangular Lagrange points L4 and L5 are linearly stable for $\mu = 3.04 \times 10^{-6}$, well below the Gascheau (Routh) stability threshold $\mu_{\text{crit}} = (1 - \sqrt{23/27})/2 \approx 0.0385$. Linearized motion about each equilibrium decomposes into two modes: a short-period epicyclic oscillation near the orbital period (≈ 365 days) and a long-period tadpole libration with $T_{\text{lib}} = T_{\text{orb}}/\sqrt{27\mu/4} \approx 220$ years. Each flanking node is placed on a tadpole orbit librating about its respective equilibrium; the spacecraft traces a small epicyclic loop that drifts slowly around the triangular point over centuries.

Because the triangular equilibria are naturally stable, no differential correction is required and the dynamics do not exhibit the exponential divergence that characterizes L1. The dominant perturbation is Jupiter's gravitational influence, which induces a secular drift requiring approximately 1.5 m/s/yr for station-keeping — sized conservatively from published operational estimates for Sun–Earth Trojan missions. Solar radiation pressure is a secondary perturbation at this mass ratio.

A.4 ΔV and Propellant Budget

Table A.4 summarizes the consolidated ΔV budget for the three-node constellation. Halo Orbit Insertion (HOI) and Tadpole Orbit Insertion (TOI) use bipropellant-powered kick stage ($I_{\text{sp}} \approx 310$ s), which is therefore jettisoned after completing its mission. All station-keeping uses xenon electric propulsion ($I_{\text{sp}} \approx 2500$ s). All propellant masses are computed via the Tsiolkovsky equation with a 20% ullage margin applied to xenon quantities. The L1 total of 125 m/s and L4/L5 total of 86 m/s are conservative relative to heritage missions.

Component	L1 ΔV (m/s)	L1 Prop. (kg)	L4/L5 ΔV (m/s)	L4/L5 Prop. (kg)
HOI/TOI (biprop, $I_{\text{sp}} \approx 310$ s)	50	15.0 (chem)	50	13.4 (chem)
Station-keeping (Xe, $I_{\text{sp}} \approx 2500$ s, 12 yr)	60	2.2 (Xe)	26	0.9 (Xe)
Contingency	15	—	10	—
Total	125	17.2	86	14.3

Table A.4 — ΔV and propellant budget for the HELIOS constellation.

References

- [1] Richardson, D. L. (1980). "Analytic Construction of Periodic Orbits About the Collinear Points." *Celestial Mechanics*, 22(3), 241–253. <https://articles.adsabs.harvard.edu/full/1980CeMec..22..241R/0000241.000.html>
- [2] Howell, K. C. (1984). "Three-Dimensional Periodic Halo Orbits." *Celestial Mechanics*, 32(1), 53–71. <https://adsabs.harvard.edu/full/1984CeMec..32...53H>
- [3] Lo, M. W., Williams, B. G., Bollman, W. E., et al. (2001). "Genesis Mission Design." https://www.researchgate.net/publication/297788197_Genesis_mission_design
- [4] Koon, W. S., Lo, M. W., Marsden, J. E. & Ross, S. D. (2011). *Dynamical Systems, the Three-Body Problem and Space Mission Design*. Caltech. https://www.cds.caltech.edu/~marsden/volume/missiondesign/KoLoMaRo_DMissionBook_2011-04-25.pdf

Figures

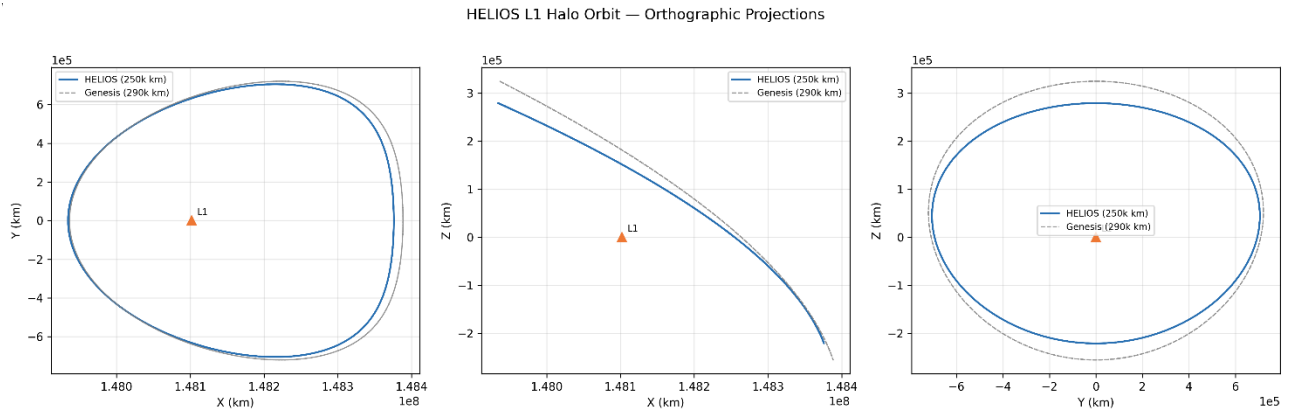


Figure A.1— Orthographic projections of the HELIOS L1 Northern halo orbit (blue, $A_z \approx 250,000$ km) and Genesis reference orbit (dashed gray, $A_z \approx 290,000$ km) in the Sun–Earth CR3BP rotating frame. Left: ecliptic (X–Y) plane. Centre: meridian (X–Z) plane. Right: view from Sun (Y–Z).

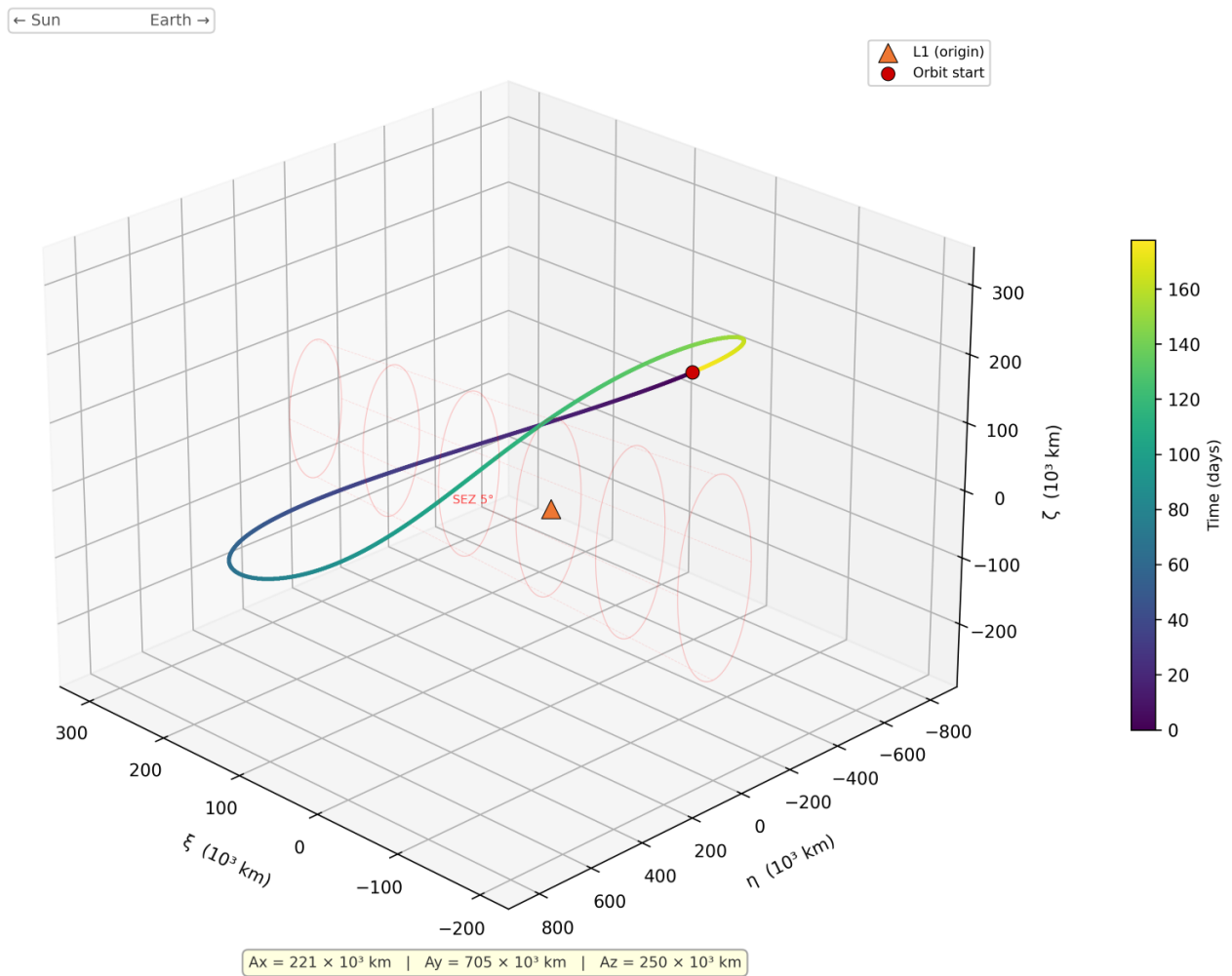


Figure A.2— Three-dimensional visualization of the HELIOS L1 Northern halo orbit (one period, $T = 177.74$ days). Coordinates are L1-centered in the rotating frame; gradient color indicates elapsed time. Red wireframe rings denote the $\pm 5^\circ$ Solar Exclusion Zone cone projected from Earth. Orbit amplitudes: $A_x \approx 221 \times 10^3$ km, $A_y \approx 705 \times 10^3$ km, $A_z \approx 250 \times 10^3$ km.

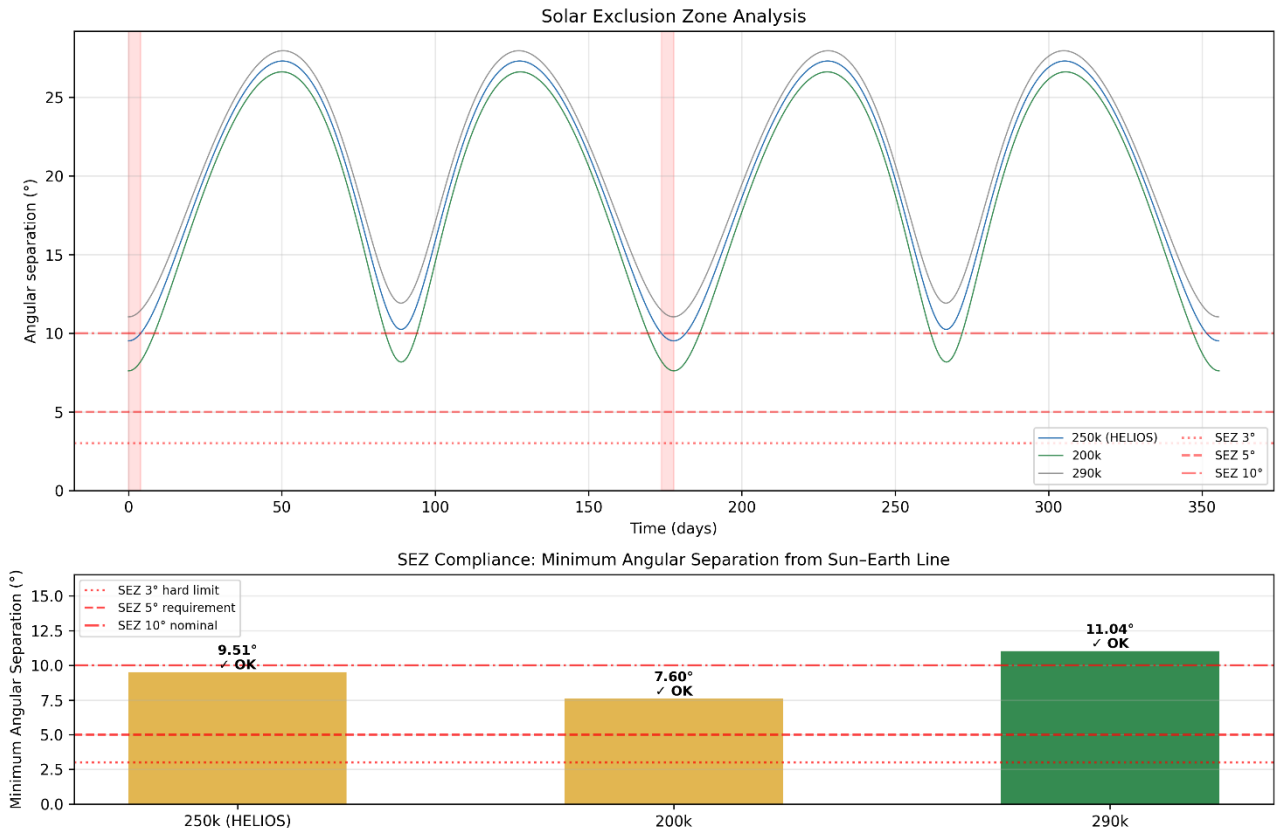


Figure A.3 — Solar Exclusion Zone compliance. Top: angular separation from the Sun–Earth line over two orbital periods for three Az amplitudes ($\approx 200\text{k}$, $\approx 250\text{k}$, $\approx 290\text{k}$ km). Bottom: minimum angular separation comparison; the HELIOS design point (250,000 km) achieves a 9.51° minimum, exceeding the 5° requirement with a 4.51° margin.

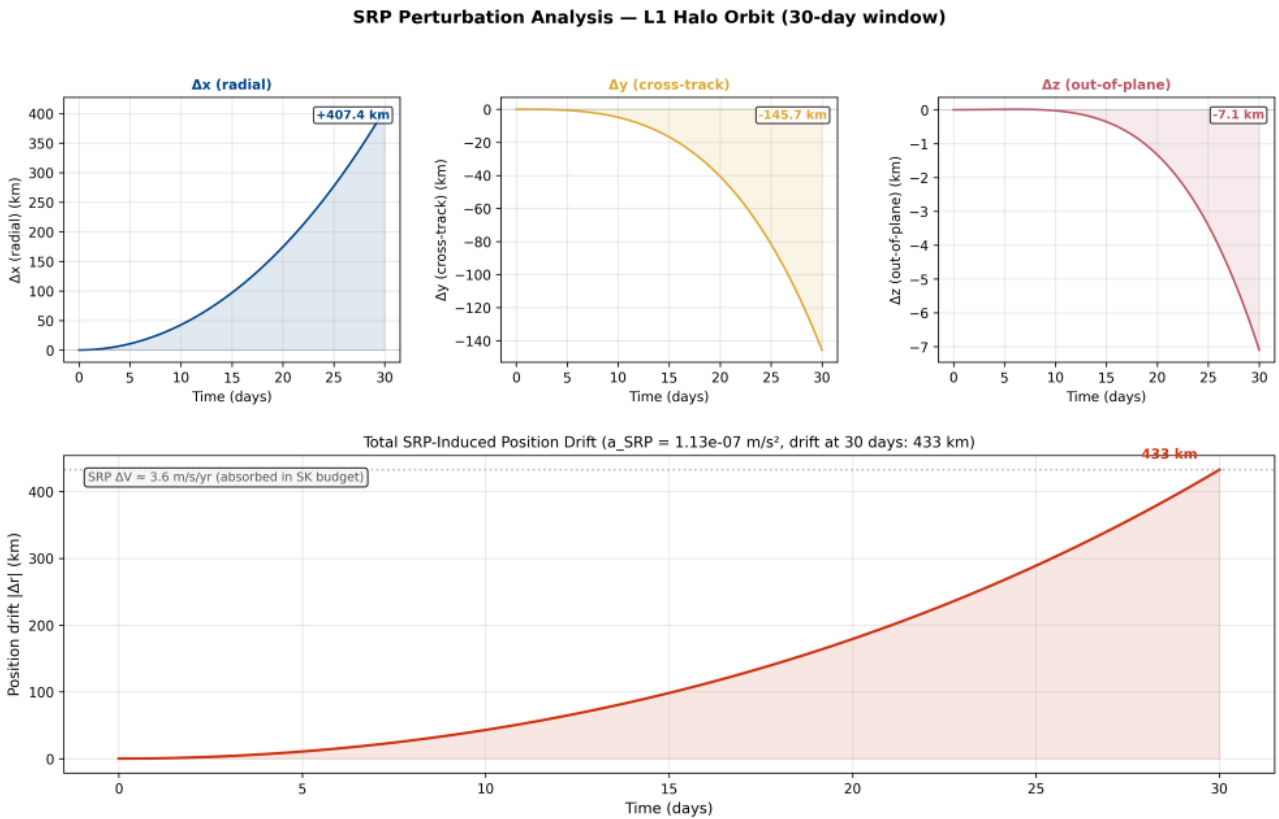


Figure A.4 — Solar Radiation Pressure perturbation residuals over a 30-day window. Top row: position drift components (radial, cross-track, out-of-plane) between CR3BP-only and CR3BP+SRP trajectories. Bottom: total drift magnitude reaching 433 km, corresponding to a 3.6 m/s/yr SRP contribution absorbed within the 60 m/s station-keeping allocation.

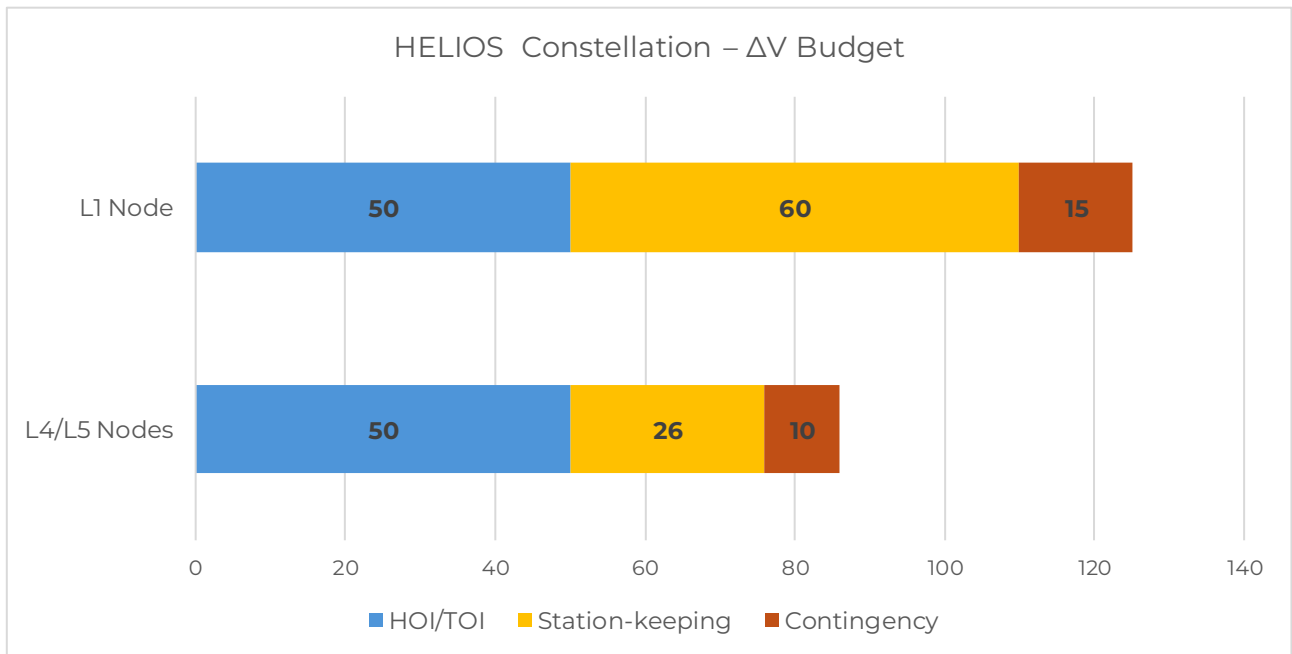


Figure A.5 — ΔV budget per node type, split between Halo Orbit Insertion (HOI), Tadpole Orbit Insertion (TOI) and Station-Keeping (SK) sized for a 12-year mission. The L1 node totals 125 m/s (50 HOI + 60 SK + 15 contingency), driven by its dynamic instability; L4/L5 nodes total 86 m/s (50 TOI + 26 SK + 10 contingency), reflecting the stable tadpole orbit's lower perturbation environment.